The second statement states that the length is 2 metres more than three times the width which is a straight line written as $l = 3b + 2$ ... (2)

Now we shall form table for the above equation (2).

b	2	4	5	8
$3b$	6	12	15	24
2	2	2	2	2
$l = 3b + 2$	8	14	17	26

Points: (2,8), (4,14), (5,17), (8,26)

The solution is the point that is common to both the lines. Here we find it to be (4,14). We can give the solution to be $b = 4$, $l = 14$.

Verification :

$$\begin{aligned} 2(l+b) &= 36 \quad ...(1) \\ 2(14+4) &= 36 \\ 2 \times 18 &= 36 \\ 36 &= 36 \quad \text{true} \end{aligned} \qquad \begin{aligned} l &= 3b + 2 \quad ...(2) \\ 14 &= 3(4) + 2 \\ 14 &= 12 + 2 \\ 14 &= 14 \quad \text{true} \end{aligned}$$

Exercise 3.10

1. Draw the graph for the following

   (i) $y = 2x$ (ii) $y = 4x - 1$ (iii) $y = \left(\frac{3}{2}\right)x + 3$ (iv) $3x + 2y = 14$

2. Solve graphically

   (i) $x + y = 7;\ x - y = 3$ (ii) $3x + 2y = 4;\ 9x + 6y - 12 = 0$

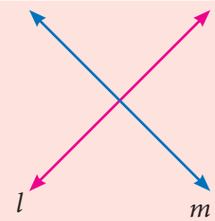
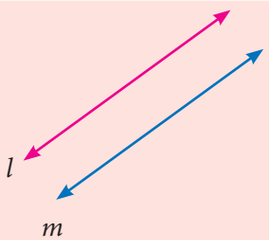
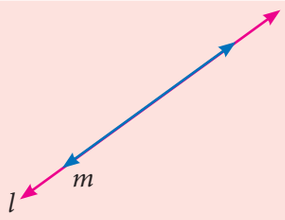
   (iii) $\frac{x}{2} + \frac{y}{4} = 1;\ \frac{x}{2} + \frac{y}{4} = 2$ (iv) $x - y = 0;\ y + 3 = 0$

   (v) $y = 2x + 1;\ y + 3x - 6 = 0$ (vi) x = –3; y = 3

3. Two cars are 100 miles apart. If they drive towards each other they will meet in 1 hour. If they drive in the same direction they will meet in 2 hours. Find their speed by using graphical method.

Some special terminology

We found that the graphs of equations within a system tell us how many solutions are there for that system. Here is a visual summary.

Intersecting lines	Parallel lines	Coinciding lines
		
One single solution	*No solution*	*Infinite number of solutions*

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